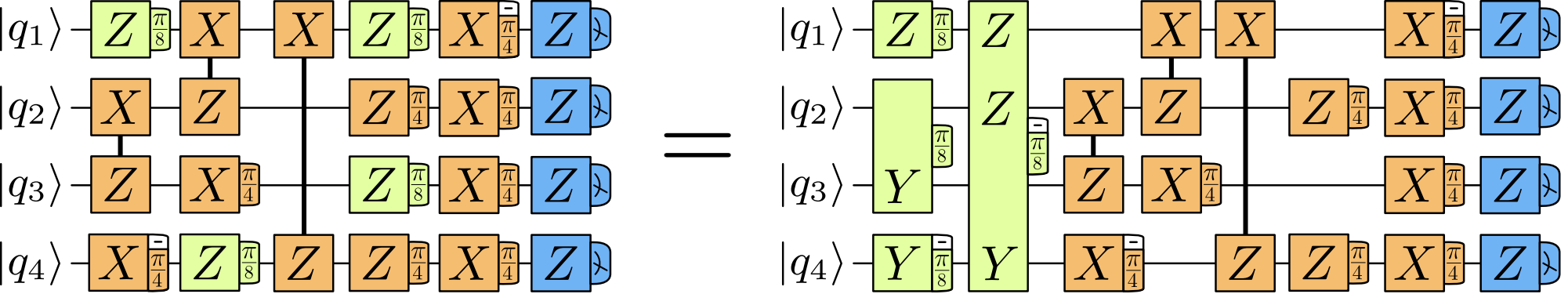
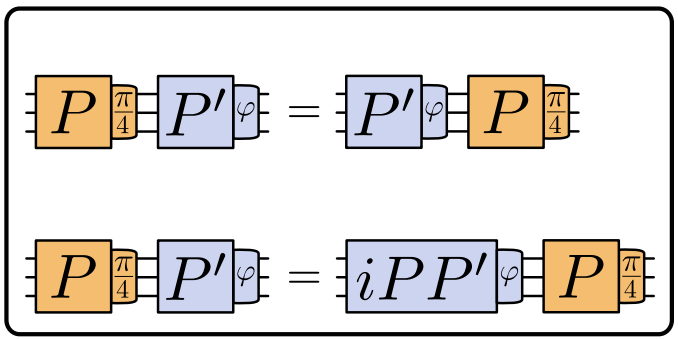
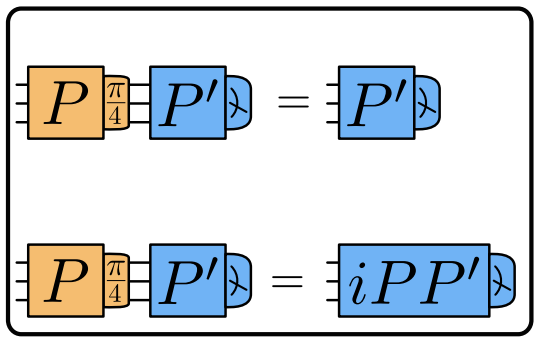
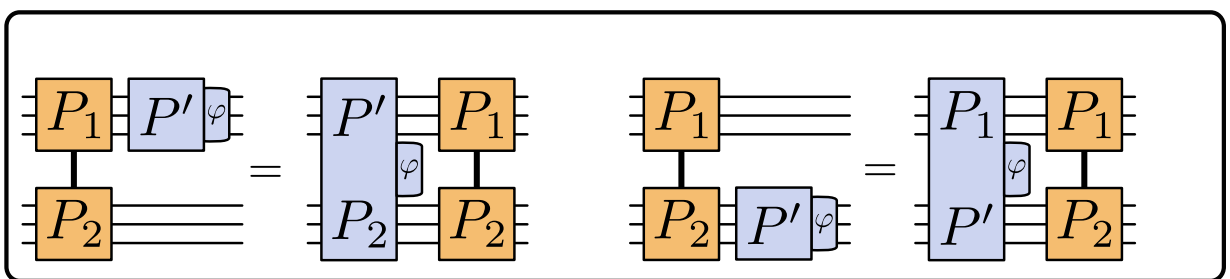
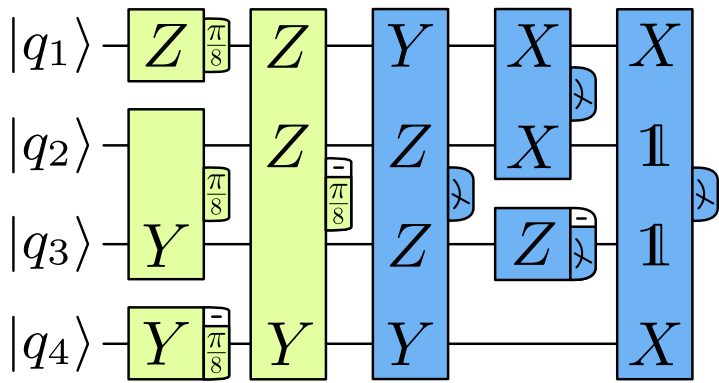
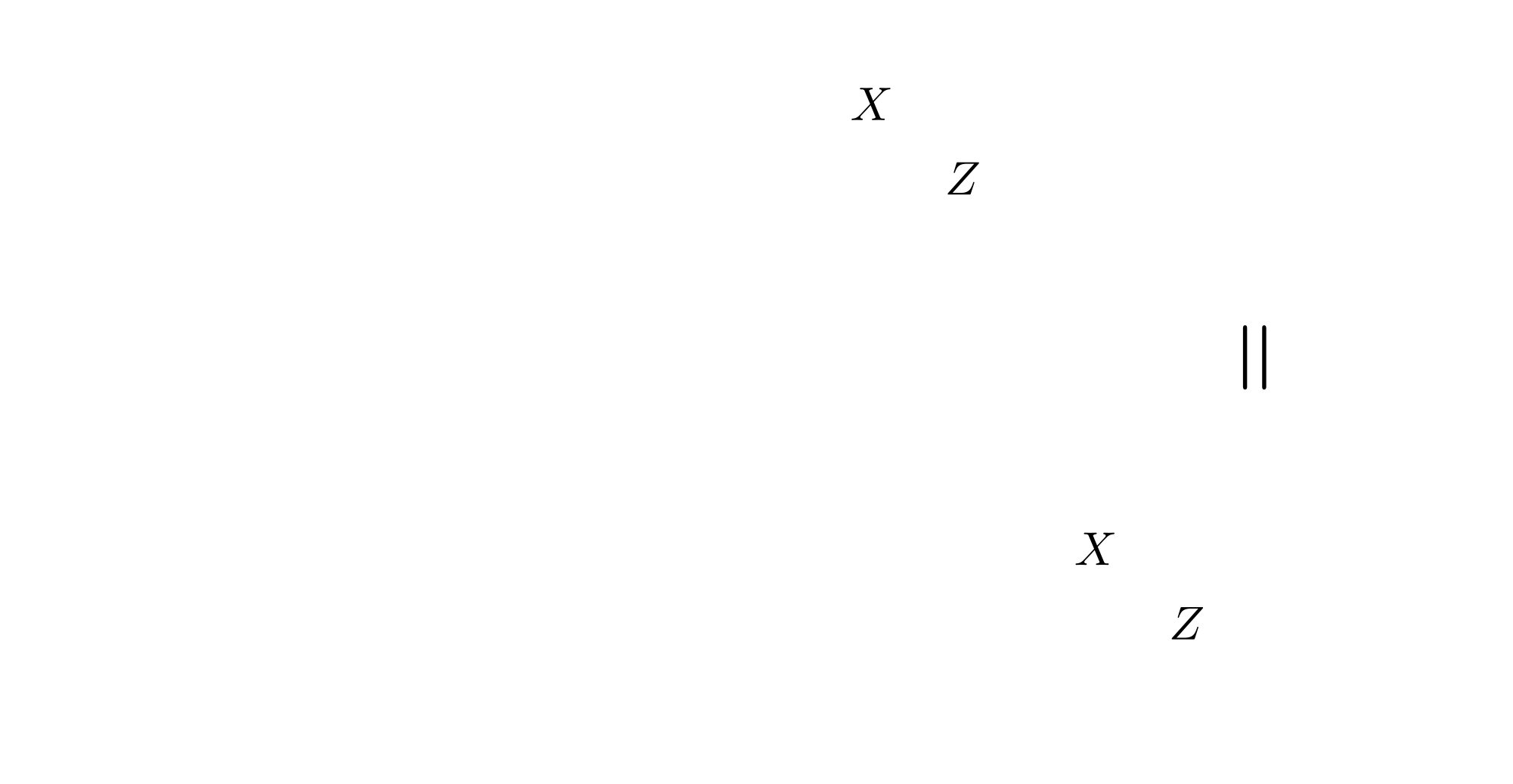

$|q_1\rangle$
$|q_2\rangle$
$|q_3\rangle$
$|q_4\rangle$
Z $\frac{\pi}{8}$ X X Z $\frac{\pi}{8}$ X $-\frac{\pi}{4}$ Z χ
X Z Z $\frac{\pi}{4}$ X $\frac{\pi}{4}$ Z χ
Z X $\frac{\pi}{4}$ Z $\frac{\pi}{8}$ X $\frac{\pi}{4}$ Z χ
X $-\frac{\pi}{4}$ Z $\frac{\pi}{8}$ Z Z $\frac{\pi}{4}$ X $\frac{\pi}{4}$ Z χ
=
$|q_1\rangle$
$|q_2\rangle$
$|q_3\rangle$
$|q_4\rangle$
Z $\frac{\pi}{8}$ Z X X X $-\frac{\pi}{4}$ Z χ
Y $\frac{\pi}{8}$ Z $-\frac{\pi}{8}$ X Z Z $\frac{\pi}{4}$ X $\frac{\pi}{4}$ Z χ
Z X $\frac{\pi}{4}$ X $\frac{\pi}{4}$ Z χ
Y $-\frac{\pi}{8}$ Y X $-\frac{\pi}{4}$ Z Z $\frac{\pi}{4}$ X $\frac{\pi}{4}$ Z χ


P $\frac{\pi}{4}$ P' φ = P' φ P $\frac{\pi}{4}$
P $\frac{\pi}{4}$ P' φ = iPP' φ P $\frac{\pi}{4}$


P $\frac{\pi}{4}$ P' χ = P' χ
P $\frac{\pi}{4}$ P' χ = iPP' χ


P_1 P' φ P_2 = P' φ P_1 P_2
P_1 P_2 P' φ = P_1 φ P_1 P' P_2


$|q_1\rangle$
$|q_2\rangle$
$|q_3\rangle$
$|q_4\rangle$
Z $\frac{\pi}{8}$ Z Y X X
Y $\frac{\pi}{8}$ Z Z Z χ
Y $-\frac{\pi}{8}$ Z Z $-\chi$
Y $-\frac{\pi}{8}$ Y Y X $\mathbb{1}$ χ
$\mathbb{1}$
X


X
Z
||
X
Z